1023 Quiz2_MCQ

1. A candy company produces individually wrapped candies. The quality control manager for the company believes that the weight of the candies is approximately normally distributed with mean 720 milligrams (mg).If the manager's belief is correct, which of the following intervals of weights will contain the largest proportion of the candies in the distribution of weights?

(A) 740 mg to 780 mg

(B) 700 mg to 740 mg ✓

(C) 680 mg to 720 mg

(D) 660 mg to 700 mg

2. The distribution of long jump distance in a statewide high school league is approximately normal with a mean of 18.5 feet and a standard deviation of 3.2 feet. Two students, Joshua and Mateo, from this league competed in the long jump. Joshua's distance was 0.52 standard deviation below the mean of the distribution. Mateo's distance was at the 48th percentile of the distribution.

At approximately what percentile was the distance jumped by Joshua, and how did his distance compare to Mateo's distance?

(A) Joshua's distance was at the 30th percentile of the distribution and was less than Mateo's distance. ✓

(B) Joshua's distance was at the 30th percentile of the distribution and was greater than Mateo's distance.

(C) Joshua's distance was at the 48th percentile of the distribution and was less than Mateo's distance.

(D) Joshua's distance was at the 70th percentile of the distribution and was greater than Mateo's distance.

Answer A

Correct. Because Joshua's distance was 0.52 standard deviation below the mean of the distribution, the probability of a distance being jumped that is less than or equal to Joshua's distance is $P(z \leq -0.52) = 0.30$. The pth percentile is the value that has $p\%$ of the distribution less than or equal to it. Therefore, the distance jumped by Joshua is the $30\,\text{th}$ percentile which is less than the distance jumped by Mateo, the $48\,\text{th}$ percentile.

3. A store owner reports that the probability that a customer who purchases a lawn mower will also purchase an extended warranty is 0.68.

Which of the following is the best interpretation of the probability 0.68 ?

(A) For all customers who purchase a lawn mower, 68% will also purchase an extended warranty. ✓

(B) For all customers of the store, 68% will purchase a lawn mower.

(C) For all customers who purchase an extended warranty, 68% will use the warranty.

(D) From the next 25 customers, 17 will purchase an extended warranty.

Answer A

Correct. In the long run, the relative frequency with which a customer who purchases a lawn mower will purchase an extended warranty is 0.68.

4. A complex electronic device contains three components, A, B, and C. The probabilities of failure for each component in any one year are 0.01, 0.03, and 0.04, respectively. If any one component fails, the device will fail. If the components fail independently of one another, what is the probability that the device will **not** fail in one year?

(A) Less than 0.01

(B) 0.080

(C) 0.922 ✓

(D) Greater than 0.99

5. Ms. Tucker travels through two intersections with traffic lights as she drives to the market. The traffic lights operate independently. The probability that both lights will be red when she reaches them is 0.22. The probability that the first light will be red and the second light will not be red is 0.33. What is the probability that the second light will be red when she reaches it?

(A) 0.40 ✓

(B) 0.45

(C) 0.50

(D) 0.55

1023 Quiz2_MCQ

6. Joe and Matthew plan to visit a bookstore. Based on their previous visits to this bookstore, the probability distributions of the number of books they will buy are given below.

Number of books Joe will buy	0	1	2
Probability	0.50	0.25	0.25

Number of books Matthew will buy	0	1	2
Probability	0.25	0.50	0.25

Assuming that Joe and Matthew make their decisions independently, what is the probability that they will purchase no books on this visit to the bookstore?

(A) 0.1250 ✓

(B) 0.1875

(C) 0.2500

(D) 0.7500

7. The table shows the responses from 103 people when asked if they support a proposal to expand the public library.

	Under the Age of 55	Age 55 or Older	Total
Yes	17	8	25
No	42	36	78
Total	59	44	103

One person from those who responded will be selected at random.

Which of the following is closest to the probability that the person selected will be someone who responded no, given that the person selected is age 55 or older?

(A) 0.350

(B) 0.462

(C) 0.757

(D) 0.818 ✓

Answer D

Correct. The condition given specifies that the person selected is age 55 or older, and this condition restricts the sample space to 44 people. Of those 44 people, 36 responded no, so the probability is found by $\frac{36}{44} \approx 0.818$.

8. Traffic data revealed that 35 percent of automobiles traveling along a portion of an interstate highway were exceeding the legal speed limit. Using highway cameras and license plate registrations, it was also determined that 52 percent of sports cars were also speeding along the same portion of the highway. What is the probability that a randomly selected car along the same portion of the highway was a speeding sports car?

(A) 0.673

(B) 0.182

(C) 0.170

(D) It cannot be determined from the information given. ✓

1023 Quiz2_MCQ

9. In a certain school, 17 percent of the students are enrolled in a psychology course, 28 percent are enrolled in a foreign language course, and 32 percent are enrolled in either a psychology course or a foreign language course or both. What is the probability that a student chosen at random from this school will be enrolled in both a foreign language course and a psychology course?

(A) 0.32

(B) 0.20

(C) 0.13 ✓

(D) 0.05

10. A random sample of 1,092 people were asked whether color was a consideration in buying a new car. They were also asked to identify one additional feature that was important. The responses are shown in the table.

	Color Consideration			
	Yes	No	Maybe	Total
Comfort	40	96	12	148
Cost	108	68	8	184
Performance	62	62	12	136
Reliability	128	116	4	248
Safety	152	192	32	376
Total	490	534	68	1,092

Which of the following is closest to the proportion of people who responded no to color consideration and who identified safety as the additional feature that was important?

(A) 0.18 ✓

(B) 0.34

(C) 0.36

(D) 0.51

Answer A

Correct. Of the 1,092 people who responded, 192 responded no to color consideration and also identified safety as the additional feature that is important. The proportion of people who responded no to color consideration and who identified safety as the additional feature that was important is $\frac{192}{1,092} \approx 0.18$.

11. Clark drives through two traffic lights on his way to work each day. From experience, he estimates that the probability that the first light will be red when he reaches it is 0.3. He also estimates that the probability that the second light will be red when he reaches it is 0.45. The lights are far enough apart that they can be considered to operate independently of each other.

Based on Clark's estimates, what is the probability that at least one of the two lights will be red on a randomly selected drive to work?

(A) 0.135

(B) 0.385

(C) 0.615 ✓

(D) 0.750

Answer C

Correct. Let F represent the event that the first light will be red on a randomly selected drive to work. Let S represent the probability that the second light will be red on a randomly selected drive to work. The probability that at least one of the two lights will be red on a randomly selected drive to work is given as $1 - \mathrm{P}\left(F^c + S^c\right)$. Because F and S can be considered independent, $1 - P\left(F^c + S^c\right) = 1 - (0.70)\,(0.55) = 0.615$.

1023 Quiz2_MCQ

12. A student is applying to two different agencies for scholarships. Based on the student's academic record, the probability that the student will be awarded a scholarship from Agency A is 0.55 and the probability that the student will be awarded a scholarship from Agency B is 0.40. Furthermore, if the student is awarded a scholarship from Agency A, the probability that the student will be awarded a scholarship from Agency B is 0.60. What is the probability that the student will be awarded at least one of the two scholarships?

(A) 0.62 ✓

(B) 0.71

(C) 0.73

(D) 0.95

Answer A

Correct. The probability that a student will be awarded at least one of the two scholarships is $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. To find $P(A \cap B)$, use the formula for conditional probability, $P\left(B \middle| A\right) = \frac{P(A \cap B)}{P(A)}$, then solve $P(A \cap B) = [P(A) \cdot P(B|A)] = (0.55)(0.6) = 0.33$. Then $P(A \cup B) = 0.55 + 0.40 - 0.33 = 0.62$.

13. A certain spinner is divided into 6 sectors of equal size, and the spinner is equally likely to land in any sector. Four of the 6 sectors are shaded, and the remaining sectors are not shaded.

Which of the following is the best interpretation of the probability that one spin of the spinner will land in a shaded sector?

(A) For many spins, the long-run relative frequency with which the spinner will land in a shaded sector is $\frac{1}{3}$.

(B) For many spins, the long-run relative frequency with which the spinner will land in a shaded sector is $\frac{1}{2}$.

(C) For many spins, the long-run relative frequency with which the spinner will land in a shaded sector is $\frac{2}{3}$. ✓

(D) For 6 spins, the spinner will land in a shaded sector 4 times.

Answer C

Correct. The spinner has 6 possible outcomes. There are 4 outcomes where the spinner will land in a shaded sector. In the long run, the relative frequency with which the spinner lands in a shaded sector is $\frac{2}{3}$.

14. Karmen and Sophia are on a swim team and race in different events at a swim meet. The probability that Karmen will win her race at the swim meet is 0.1. The probability that Sophia will win her race at the swim meet is 0.3.

If the events of Karmen winning her race and Sophia winning her race are independent, what is the probability that either Karmen or Sophia will win her race at the swim meet?

(A) 0.03

(B) 0.30

(C) 0.37 ✓

(D) 0.40

Answer C

Correct. The $P(\text{Karmen wins} \cup \text{Sophia wins}) = P(\text{Karmen wins}) + P(\text{Sophia wins}) - P(\text{Karmen wins} \cap \text{Sophia wins})$. Because the events of Karmen winning her race and Sophia winning her race are independent, $P(K\text{armen wins} \cap \text{Sophia wins}) = P(\text{Karmen wins}) \times P(\text{Sophia wins})$. Therefore, the $P(\text{Karmen wins} \cup \text{Sophia wins}) = P(\text{Karmen wins}) + P(\text{Sophia wins}) - P(\text{Karmen wins}) \times P(\text{Sophia wins}) = 0.1 + 0.3 - 0.1 \times 0.3 = 0.37$

1023 Quiz2_MCQ

15. As part of a survey, 118 adults with tattoos were asked whether they got their first tattoo before reaching the age of 25 and whether their first tattoo was done by a professional tattoo artist. The resulting data are summarized in the following table.

	Before the Age of 25	**25 or Older**	**Total**
Professional tattoo artist	10	87	97
Not a professional tattoo artist	9	12	21
Total	19	99	118

Which of the following provides evidence that age at first tattoo and whether the first tattoo was done by a professional tattoo artist are not independent for the group of 118 adults with tattoos?

(A) $\frac{10}{97} \neq \frac{87}{97}$

(B) $\frac{10}{97} \neq \frac{19}{118}$ ✓

(C) $\frac{10}{19} \neq \frac{10}{97}$

(D) $\frac{97}{118} \neq \frac{21}{118}$

Answer B

Correct. If, and only if, events A and B are independent, then $P(A|B) = P(A)$, $P(B|A) = P(B)$, and $P(A \cap B) = P(A) \cdot P(B)$. $P(\text{Before the Age of 25}) = \frac{19}{118}$. $P(\text{Before the Age of 25}|\text{Professional tattoo artist}) = \frac{10}{97}$. Because $P(\text{Before the Age of 25}|\text{Professional tattoo artist})$ does not equal $P(\text{Before the Age of 25})$, the events Before the Age of 25 and Professional tattoo artist are not independent.

16. The distribution of monthly rent for one-bedroom apartments in a city is approximately normal with mean $936 and standard deviation 61. A graduate student is looking for a one-bedroom apartment and wants to pay no more than $800 in monthly rent. Of the following, which is the best estimate of the percent of one-bedroom apartments in the city with a monthly rent of at most $800 ?

(A) 1.3% ✓

(B) 2.5%

(C) 50%

(D) 95%

17. The commuting time for a student to travel from home to a college campus is normally distributed with a mean of 30 minutes and a standard deviation of 5 minutes. If the student leaves home at 8:25 A.M., what is the probability that the student will arrive at the college campus later than 9 A.M.?

(A) 0.16 ✓

(B) 0.32

(C) 0.84

(D) 1.00

18. At a certain store, the distribution of weights of cartons of large eggs is approximately normal with mean 26 ounces (oz). Based on the distribution, which of the following intervals will contain the greatest proportion of cartons of large eggs at the store?

(A) 22 oz to 26 oz

(B) 24 oz to 28 oz ✓

(C) 26 oz to 30 oz

(D) 28 oz to 32 oz

19. For a certain population of penguins, the distribution of weight is approximately normal with mean 15.1 kilograms (kg) and standard deviation 2.2 kg. Approximately what percent of the penguins from the population have a weight between 13.0 kg and 16.5 kg?

(A) 34%

(B) 57% ✓

(C) 68%

(D) 74%

1023 Quiz2_MCQ

20. The distribution of lengths of salmon from a certain river is approximately normal with standard deviation 3.5 inches. If 10 percent of salmon are longer than 30 inches, which of the following is closest to the mean of the distribution?

(A) 26 inches ✓

(B) 28 inches

(C) 30 inches

(D) 34 inches

Answer A

Correct. Since 10% of the salmon are longer than 30 inches, then 90% of the salmon are less than or equal to 30 inches. From the standard normal table, approximately 90% of the area under the curve falls at or below the z-score of 1.282, so this value can be substituted into the formula for a z-score $\left(z = \frac{x-\mu}{\sigma} = \frac{30-\mu}{3.5} = 1.282\right)$ and then solved for μ.